

SafeExec Draft Report, Deck, and Final Test Evidence

AI-use disclosure. Drafted with Codex (GPT-5, Codex desktop app) from repository artifacts and fresh W12 evidence. AI-drafted, student-revised.
Full audit trail: `docs/ai-use-log.md`.

Course: CISC 699 Applied Project in Computer Information Sciences

Student: Zixuan Liang

Advisor: Prof. Khalid Lateef

Instructor: Dr. Majid Shaalan

Assignment: 12 Hard Stop 6: Draft Report, Deck, and Final Test Evidence

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1. Package Purpose

This package connects the three required streams for Hard Stop 6:

1. a near-final technical report draft,
2. a presentation/demo deck and walkthrough script, and
3. final-test evidence that supports the report and deck claims.

The main story is consistent across all three: SafeExec has moved from a scaffold into a reproducible, testable sandbox artifact. The current evidence supports the claim that the API/result path, local validation, reproducibility audit, and Docker/gVisor benchmark harness are functioning. The remaining work is corpus depth and final benchmark refresh, not a change in project direction.

2. Current Draft Report Status

Primary draft report file:

`near-final-technical-report-draft.md`

The draft now includes:

- abstract and problem statement,
- related-work synthesis,
- approved scope and requirements,
- system design and implementation status,

- evaluation methodology,
- current results and interpretation,
- ethics/security/privacy discussion,
- limitations and final-week work plan,
- reference list aligned to the W3 literature brief.

This draft is not yet the W14 final report. The remaining report work is to insert the final HumanEval/MBPP subset table, expanded adversarial results, final Docker/gVisor benchmark tables, and final demo screenshots or transcript.

3. Presentation and Demo Readiness

Primary deck:

SafeExec-Draft-Report-and-Demo-Deck.pptx

Primary walkthrough script:

walkthrough-script.md

The deck follows the same report arc:

1. problem and project claim,
2. artifact architecture,
3. current implementation,
4. evidence collected,
5. Docker/gVisor findings,
6. limitations and remaining work,
7. final-week plan.

The deck intentionally avoids adding claims that do not appear in the report.

Every numeric claim in the deck is traceable to W8, W10, or W12 evidence files.

4. Final Test Evidence Summary

Fresh W12 evidence was generated with:

```
make final-evidence
```

Summary:

Evidence stream	Result	File
Smoke execution	PASS	evidence/smoke-output.txt
Unit/functional tests	PASS	evidence/unit-tests-output.txt

Local validation/API trace	PASS	evidence/validation-summary.txt
Reproducibility audit	PASS	evidence/reproducibility-audit.md
Environment snapshot	Captured	evidence/environment-snapshot.txt

The local validation workflow passed 12/12 local records and the API trace.

The reproducibility audit passed with no blocking findings.

5. Evidence Brought Forward

The W12 folder includes copies of important prior evidence so the reviewer can see the full evidence chain in one package:

- W8 target-host Docker/gVisor summary:

evidence/w8-target-midpoint-summary.md

- W8 target-host CSV summary:

evidence/w8-target-midpoint-summary.csv

- W8 runsc version:

evidence/w8-target-runsc-version.txt

- W10 clean package run:

evidence/w10-clean-run-output.txt

The most important target-host result remains:

Backend	Passed / Total	Pass rate
Local subprocess	30 / 30	100.0%
Hardened Docker	50 / 50	100.0%
gVisor (runsc)	49 / 50	98.0%

6. Interpretation and Midpoint-to-Final Adjustment

The evidence supports the artifact's basic technical maturity:

- the API/result schema is stable enough for repeatable validation;
- Docker hardening is exercised on a real target host;
- gVisor is integrated through runsc;
- local validation and reproducibility checks are repeatable;
- the clean-run package path is documented and tested.

The evidence does not yet justify final headline claims about the complete functional corpus or adversarial containment thresholds. Those are final-week targets. The report and deck therefore present the current results as a strong engineering baseline plus a clear final evidence plan.

7. Revision Responsiveness

Earlier implementation-sprint feedback criticized missing verifiable evidence.

This package responds directly:

- evidence files are included rather than merely referenced;
- command transcripts are in evidence/;
- environment snapshots are submitted;
- reproducibility audit and clean-run evidence are included;
- report/deck/checklist all point to the same files.

8. Submission Contents

Recommended Canvas upload:

- Draft-Report-Deck-Final-Test-Evidence.docx
- Draft-Report-Deck-Final-Test-Evidence.pdf
- near-final-technical-report-draft.md
- final-test-evidence-appendix.md
- SafeExec-Draft-Report-and-Demo-Deck.pptx
- walkthrough-script.md
- canvas-submission-checklist.md
- the full evidence/ folder or a zip of it
- docs/ai-use-log.md

9. Remaining Work Before Final Submission

1. Finish HumanEval/MBPP subset manifest and license/provenance table.
2. Expand adversarial suite to the approved count/category target.
3. Re-run Docker/gVisor benchmark on the Ubuntu target host.
4. Separate cold-start and warm-run timing in final performance tables.
5. Add final results to the report and deck.
6. Record the final walkthrough/demo.
7. Rebuild final reproducibility package and rerun `make final-evidence`.